

A Controlled Study of Pure Monte Carlo Tree Search for Symbolic Regression

After conversations with GPT

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1 A controlled study of pure Monte Carlo tree search for symbolic regression

1.1 Motivation and scope

The purpose of this study is to characterize the finite-budget limitations of pure Monte Carlo tree search (MCTS) in grammar-based symbolic regression. We deliberately exclude neural policy and value networks. The neural-guidance question is scientifically separate: before asking whether a network improves search, one must determine what information the underlying pure-search procedure estimates and when that information is aligned with symbolic optimization.

The central distinction is between the quality of a typical random completion and the quality of the best reachable completion. Let s be a partial derivation and let $\mathcal{E}(s)$ denote the set of terminal expressions reachable from s . The best-achievable value is

$$V^*(s) = \max_{\pi \in \mathcal{E}(s)} R(\pi), \quad (1)$$

where $R(\pi)$ is the terminal symbolic-regression reward. By contrast, consider the fixed random-completion policy that, at each nonterminal state s , selects uniformly among the legal actions $\mathcal{A}(s)$,

$$q(a \mid s) = \frac{1}{|\mathcal{A}(s)|}, \quad a \in \mathcal{A}(s).$$

Because transitions are deterministic, following this policy from s to termination induces a distribution $q(\pi \mid s)$ over the completions $\pi \in \mathcal{E}(s)$, and the mean-rollout value is

$$V^q(s) = \mathbb{E}_{\pi \sim q(\cdot \mid s)} [R(\pi)]. \quad (2)$$

The optimization objective depends on V^* , whereas conventional UCT uses empirical mean returns as its exploitation statistic [8]. Asymptotic consistency does not by itself imply effective finite-budget search, and adverse finite-depth behavior is known for UCT [3]. This distinction is particularly relevant to symbolic regression, where the goal is to discover at least one exceptional expression rather than a policy with high average return. Best-focused Monte Carlo methods have a history in single-player search and expression discovery [1, 2], while risk-seeking symbolic regression explicitly optimizes upper-tail rather than expected performance [10]. Recent MCTS-based symbolic-regression systems also emphasize learned proposal mechanisms or extreme-value allocation rather than pure mean-based UCT [6, 7].

We isolate two potential sources of difficulty:

1. *value-semantic failure*: V^q ranks competing branches in the opposite order from V^* ; formally, for two sibling states s_1 and s_2 , a value-semantic failure exists if:

$$V^q(s_1) > V^q(s_2) \quad \text{while} \quad V^*(s_1) < V^*(s_2)$$

In this scenario, MCTS confidently exploits the wrong branch because a shallow, suboptimal "decoy" yields a high expected reward, while the branch containing the true global optimum is penalized by a low expected reward due to the abundance of poor completions.

2. *rare-event failure*: near-optimal completions have very small probability under the rollout policy. For a tolerance $\varepsilon \geq 0$, this is the probability that a random rollout yields an expression within ε of the true optimum:

$$\rho_\varepsilon(s) = \Pr_{\pi \sim q(\cdot|s)} [R(\pi) \geq V^*(s) - \varepsilon]$$

Table 1 summarizes the two mechanisms.

Failure mechanism	Formal condition	Intuitive cause	Implication for search
Value-semantic failure	$V^q(s_1) > V^q(s_2)$ while $V^*(s_1) < V^*(s_2)$ for sibling states s_1, s_2	A suboptimal branch has the higher mean rollout reward: a shallow decoy looks better on average than the branch holding the optimum.	Search confidently exploits the wrong branch and starves the branch containing the true global optimum.
Rare-event failure	$\rho_\varepsilon(s) \ll B^{-1}$	Near-optimal completions carry a negligible fraction of the rollout probability mass.	Search explores the correct branch but exhausts the budget B before sampling a near-optimal leaf.

Table 1: The two finite-budget failure mechanisms of mean-based MCTS.

Experiment 1 measures these two quantities exactly. Experiment 2 tests whether they predict the finite-budget behavior of pure MCTS.

Our goal is then to answer:

Question 1.1. After identifying the difficulty in pure MCTS, how does neural net address these?

1.2 Empirical function space

Let

$$x_1, \dots, x_n \in [-1, 1], \quad n = 41,$$

be equally spaced sample points, and define the empirical inner product

$$\langle f, g \rangle_n = \frac{1}{n} \sum_{i=1}^n f(x_i)g(x_i). \quad (3)$$

Starting from the monomials $x, x^2, \dots, x^{12}$, remove their empirical means and apply Gram–Schmidt orthogonalization,

$$\psi_1, \dots, \psi_{12}$$

satisfying

$$\langle \psi_j, 1 \rangle_n = 0, \quad \langle \psi_j, \psi_\ell \rangle_n = \delta_{j\ell}. \quad (4)$$

The first two basis functions are the normalized linear and centered-quadratic components,

$$\psi_1(x) = \frac{x}{\sqrt{\mu_2}}, \quad \psi_2(x) = \frac{x^2 - \mu_2}{\sqrt{\nu}}, \quad (5)$$

where

$$\mu_2 = \langle x, x \rangle_n, \quad \nu = \langle x^2 - \mu_2, x^2 - \mu_2 \rangle_n.$$

Thus the construction extends the orthogonal linear-quadratic decomposition used in the motivating quadratic example.

1.3 Controlled target family

For $\lambda \in (0, 1)$, define the four-component target

$$y_\lambda(x) = \sqrt{\lambda} \psi_1(x) + \sqrt{\frac{1-\lambda}{3}} (\psi_2(x) + \psi_3(x) + \psi_4(x)). \quad (6)$$

Proposition 1.2 (Unit target variance). For every $\lambda \in (0, 1)$,

$$\bar{y}_\lambda = \langle y_\lambda, 1 \rangle_n = 0, \quad \|y_\lambda\|_n^2 = \langle y_\lambda, y_\lambda \rangle_n = 1. \quad (7)$$

Proof. Both identities use only the three properties of the basis (4): each ψ_j is centered, $\langle \psi_j, 1 \rangle_n = 0$; has unit length, $\langle \psi_j, \psi_j \rangle_n = 1$; and is orthogonal to the others, $\langle \psi_j, \psi_\ell \rangle_n = 0$ for $j \neq \ell$. For the mean, linearity gives

$$\langle y_\lambda, 1 \rangle_n = \sqrt{\lambda} \langle \psi_1, 1 \rangle_n + \sqrt{\frac{1-\lambda}{3}} \sum_{m=2}^4 \langle \psi_m, 1 \rangle_n = 0.$$

Since the mean is zero, $\|y_\lambda\|_n^2$ is the empirical variance of y_λ . Expanding $\langle y_\lambda, y_\lambda \rangle_n$ produces sixteen terms; orthogonality annihilates every cross-term, and unit length reduces each surviving square to its coefficient:

$$\|y_\lambda\|_n^2 = \lambda + 3 \cdot \frac{1-\lambda}{3} = 1.$$

The identity is the Pythagorean theorem in the geometry (3). □

The parameter λ is therefore exactly the fraction of target variance carried by ψ_1 . We use

$$\lambda_{\text{weak}} = \frac{1}{4}, \quad \lambda_{\text{strong}} = 0.997. \quad (8)$$

When $\lambda = 1/4$, all four target components have equal variance share. When $\lambda = 0.997$, the first component explains 99.7% of the target variance and each remaining component explains 0.1%. The latter is a controlled analogue of a pseudo-linear target: a shallow one-term expression is almost perfectly predictive but is not structurally exact.

1.4 Controlled derivation grammar

The root state s_0 has two legal actions.

Decoy action. The first action, denoted D , terminates immediately with the shallow model class

$$\mathcal{M}_D = \text{span}\{1, \psi_1\}. \quad (9)$$

The branch contains a single terminal expression, fitted by least squares and scored by the same unpenalized R^2 criterion as every terminal expression, cf. (16) below. Every visit therefore returns one deterministic reward, and that reward is exactly the decoy parameter.

Proposition 1.3 (Decoy reward). For every $\lambda \in (0, 1)$, the least-squares fit of y_λ in $\mathcal{M}_D$ is

$$\hat{y}_D = \sqrt{\lambda} \psi_1, \quad (10)$$

and its reward is

$$R(D) = \lambda. \quad (11)$$

Proof. Because the numerator and denominator of (16) carry the same factor n , the ratio is unchanged if both are computed in the geometry (3); we do so throughout. The constant function satisfies $\langle 1, 1 \rangle_n = 1$, so by (4) the set $\{1, \psi_1\}$ is an orthonormal basis of $\mathcal{M}_D$, and the least-squares fit is the orthogonal projection

$$\hat{y}_D = \langle y_\lambda, 1 \rangle_n 1 + \langle y_\lambda, \psi_1 \rangle_n \psi_1.$$

The intercept vanishes, $\langle y_\lambda, 1 \rangle_n = 0$, by Proposition 1.2. For the slope, expand y_λ by (6); orthogonality annihilates the three cross-terms and unit length leaves

$$\langle y_\lambda, \psi_1 \rangle_n = \sqrt{\lambda} \langle \psi_1, \psi_1 \rangle_n = \sqrt{\lambda},$$

which proves (10). The residual is then explicit,

$$y_\lambda - \hat{y}_D = \sqrt{\frac{1-\lambda}{3}} (\psi_2 + \psi_3 + \psi_4),$$

and the Pythagorean theorem in the geometry (3) gives $\|y_\lambda - \hat{y}_D\|_n^2 = 3 \cdot \frac{1-\lambda}{3} = 1 - \lambda$. Since $\bar{y}_\lambda = 0$ and $\|y_\lambda\|_n^2 = 1$ (Proposition 1.2), the reward (16) evaluates to

$$R(D) = 1 - \frac{\|y_\lambda - \hat{y}_D\|_n^2}{\|y_\lambda\|_n^2} = 1 - (1 - \lambda) = \lambda. \quad \square$$

Remark 1.4 (The decoy is an exact trap). Proposition 1.3 is classical: on orthonormal regressors, least squares projects coordinatewise and R^2 is the captured share of target variance [5, §3.2.3]. Its role here is structural. At $\lambda_{\text{strong}} = 0.997$, a structurally incomplete one-term model earns a deterministic reward of 0.997 in exact arithmetic, while the sibling branch containing the exact expression returns the far smaller mean (20) under random completion. Objectives whose locally averaged signal points away from the global optimum are called *deceptive* in evolutionary computation [4]; the analogous hazard for sampling-based tree search is the shallow *search trap* [11], on which UCT’s transitory regret can grow doubly exponentially in the depth [3]. The decoy transplants this structure into single-player symbolic regression, where a near-perfect R^2 on a wrong functional form is precisely the failure mode that motivates reporting predictive-accuracy and symbolic solutions separately in benchmark evaluation [9].

Continue action. The second action, denoted C , opens the compositional branch. It contains the structurally exact four-component expression, whose reward is 1; a uniform random completion, however, recovers that expression only rarely, and the mean completion reward is p_M , cf. (20).

The action creates four atom slots $A_1, \dots, A_4$. Each slot has the productions

$$A_r \longrightarrow \psi_1 \mid \psi_2 \mid \dots \mid \psi_M, \quad r = 1, \dots, 4, \quad (12)$$

where

$$M \in \{4, 12\}. \quad (13)$$

A terminal continuation is an ordered tuple

$$\mathbf{j} = (j_1, j_2, j_3, j_4) \in \{1, \dots, M\}^4.$$

Let

$$J(\mathbf{j}) = \{j_1, j_2, j_3, j_4\}$$

be the set of distinct selected atoms. The corresponding fitted model space is

$$\mathcal{M}_{\mathbf{j}} = \text{span}(\{1\} \cup \{\psi_j : j \in J(\mathbf{j})\}). \quad (14)$$

Repeated atoms are removed before fitting, or equivalently handled by the Moore–Penrose pseudoinverse.

For a design matrix $X_{\mathbf{j}}$, the fitted coefficients are

$$\hat{a}_{\mathbf{j}} = (X_{\mathbf{j}}^T X_{\mathbf{j}})^{\dagger} X_{\mathbf{j}}^T y_{\lambda}. \quad (15)$$

The terminal reward is unpenalized R^2 ,

$$R(\mathbf{j}) = 1 - \frac{\|y_{\lambda} - X_{\mathbf{j}} \hat{a}_{\mathbf{j}}\|_2^2}{\|y_{\lambda} - \bar{y}_{\lambda} \mathbf{1}\|_2^2}. \quad (16)$$

No reward clipping or nonlinear optimizer is used.

Repeating the projection argument in the proof of Proposition 1.3, now for the orthonormal basis of $\mathcal{M}_{\mathbf{j}}$, shows that the reward has the exact form

$$R(\mathbf{j}) = \lambda \mathbf{1}\{1 \in J(\mathbf{j})\} + \frac{1 - \lambda}{3} \sum_{m=2}^4 \mathbf{1}\{m \in J(\mathbf{j})\}. \quad (17)$$

Consequently, all rewards lie in $[0, 1]$, distractor atoms $\psi_5, \dots, \psi_M$ capture no target variance, and the exact reward is independent of numerical fitting pathologies.

1.5 Exact completion statistics

Let D and C denote the root children corresponding to the decoy and continue actions. Since the decoy child admits the single completion of Proposition 1.3, it satisfies

$$V^*(D) = V^q(D) = \lambda. \quad (18)$$

Proposition 1.5 (Continue-branch values). The continue branch selects four atoms uniformly and independently from the vocabulary $\{\psi_1, \dots, \psi_M\}$. Its best-achievable reward is

$$V^*(C) = 1, \quad (19)$$

and its mean-rollout reward is

$$V^q(C) = p_M = 1 - \left(1 - \frac{1}{M}\right)^4. \quad (20)$$

Proof. The reward (17) is maximal when $J(\mathbf{j}) \supseteq \{1, 2, 3, 4\}$, where every indicator equals one and

$$R(\mathbf{j}) = \lambda + \frac{1 - \lambda}{3} \cdot 3 = 1;$$

the tuple $\mathbf{j} = (1, 2, 3, 4)$ achieves this, proving (19). For the mean, take expectations in (17) under the uniform completion policy q :

$$V^q(C) = \lambda \Pr_q(1 \in J(\mathbf{j})) + \frac{1 - \lambda}{3} \sum_{m=2}^4 \Pr_q(m \in J(\mathbf{j})).$$

A fixed atom m is missed by one slot with probability $1 - \frac{1}{M}$, hence by all four independent slots with probability $(1 - \frac{1}{M})^4$, so $\Pr_q(m \in J(\mathbf{j})) = p_M$ for every m . Substituting gives $V^q(C) = \lambda p_M + (1 - \lambda) p_M = p_M$. $\square$

Comparing (18) with (20), the mean-rollout evaluator reverses the correct root ordering exactly when

$$\lambda > p_M. \quad (21)$$

Exact recovery requires the four selected atoms to form a permutation of $(1, 2, 3, 4)$. Among the M^4 equally likely ordered tuples, exactly $4!$ are permutations of the target, hence

$$\rho_0(C) = \Pr_q(R = 1 \mid C) = \frac{4!}{M^4}. \quad (22)$$

Cell	λ	M	$V^q(C)$	$\rho_0(C)$	Inversion	$K_{0.95}$
Weak decoy, low distraction	0.25	4	0.683594	0.09375	No	31
Strong decoy, low distraction	0.997	4	0.683594	0.09375	Yes	31
Weak decoy, high distraction	0.25	12	0.293933	0.001157	No	2587
Strong decoy, high distraction	0.997	12	0.293933	0.001157	Yes	2587

Table 2: Controlled 2×2 design. The decoy parameter λ controls whether the random-completion mean has the correct root ordering, while M controls the mass of exact completions. The column $K_{0.95}$ is conditional on repeatedly sampling the continue branch.

Question 1.6. What is the minimum number of rollouts (K) we must perform to guarantee, with 95% confidence, that we will see the exact solution at least once?

Proposition 1.7 (Hit probability and sample complexity). Draw K independent uniform completions of C . The probability that at least one is exact is

$$P_{\text{hit}}(K) = 1 - (1 - \rho_0(C))^K, \quad (23)$$

and, for $\delta \in (0, 1)$, the guarantee $P_{\text{hit}}(K) \geq 1 - \delta$ holds if and only if

$$K \geq K_{1-\delta} = \left\lceil \frac{\log \delta}{\log(1 - \rho_0(C))} \right\rceil. \quad (24)$$

Proof. A single completion misses the exact expression with probability $1 - \rho_0(C)$. By independence, all K completions miss with probability $(1 - \rho_0(C))^K$, and the complement of that event is (23). The guarantee $P_{\text{hit}}(K) \geq 1 - \delta$ is therefore equivalent to

$$(1 - \rho_0(C))^K \leq \delta.$$

Taking logarithms gives $K \log(1 - \rho_0(C)) \leq \log \delta$, and dividing by $\log(1 - \rho_0(C))$, which is negative, reverses the inequality:

$$K \geq \frac{\log \delta}{\log(1 - \rho_0(C))}.$$

The smallest integer satisfying this bound is the ceiling in (24). $\square$

Table 2 gives the four environments.

2 Experiment 1: exact value-semantic audit

Question. Does mean random-completion return preserve the ordering induced by best-achievable reward, and how rare is the exact expression within each partial derivation?

Hypothesis 1. The rollout mean ranks the continue branch above the decoy for $\lambda = 0.25$ and below the decoy for $\lambda = 0.997$:

$$V^q(C) > \lambda_{\text{weak}}, \quad V^q(C) < \lambda_{\text{strong}}. \quad (25)$$

Independently, the completion budget required for exact recovery scales as

$$K_{1-\delta} = \Theta\left(\frac{\log(1/\delta)}{\rho_0}\right) \quad \text{as } \rho_0 \rightarrow 0. \quad (26)$$

Procedure. No stochastic tree search is used in this experiment. For each of the four cells, enumerate all M^4 terminal continuations and all partial prefixes of depths 0, 1, 2, 3, 4. For each partial state s , compute exactly

$$V^*(s), \quad V^q(s), \quad \rho_0(s). \quad (27)$$

For a nonterminal state s , define the action selected by the mean-rollout oracle as

$$a_q(s) = \arg \max_a V^q(T(s, a)), \quad (28)$$

and define its oracle sibling regret by

$$g_q(s) = V^*(s) - V^*(T(s, a_q(s))). \quad (29)$$

The largest environment contains only

$$1 + 12 + 12^2 + 12^3 + 12^4 = 22,621$$

continue-prefix states, so exhaustive calculation is tractable.

The implementation must compute each terminal reward in two ways:

1. using the analytical expression (17);
2. using the actual least-squares evaluator (15)–(16).

The two values must agree to numerical tolerance 10^{-12} .

Primary outputs. Report the exact root quantities

$$V^*(D), \quad V^q(D), \quad V^*(C), \quad V^q(C), \quad \rho_0(C), \quad K_{0.95}.$$

Plot:

1. the terminal completion-reward distribution in each cell;
2. $V^q(s)$ against $V^*(s)$, stratified by state depth;
3. sibling regret $g_q(s)$ against depth;
4. the exact hit curve $1 - (1 - \rho_0)^K$.

Because Experiment 1 is an exact census, it requires no random seeds, confidence intervals, or significance tests. Failure to reproduce the analytical identities indicates an implementation error and should prevent proceeding to Experiment 2.

3 Experiment 2: finite-budget pure-MCTS limits

Question. At matched terminal-evaluation budgets, do the value-order mismatch and exact-completion rarity identified in Experiment 1 predict the behavior of actual pure search?

3.0.1 Search conditions

We compare three conditions.

Uniform random search. Sample complete derivations independently from the uniform grammar policy and return the best terminal expression observed.

Mean-UCT. At node s , select

$$a_t = \arg \max_a \left[\bar{Q}_t(s, a) + \sqrt{\frac{2 \log N_t(s)}{N_t(s, a)}} \right], \quad (30)$$

where $\bar{Q}_t(s, a)$ is the empirical mean of terminal returns from simulations passing through (s, a) . Unvisited actions receive infinite score. Each simulation selects through the current tree, expands one unvisited edge, performs one uniform random completion, and backs up the terminal reward by an arithmetic mean.

Max-UCT diagnostic. Define

$$Q_t^{\max}(s, a) = \max \{R(\pi_b) : \pi_b \text{ was sampled through } (s, a)\}. \quad (31)$$

Select using

$$a_t = \arg \max_a \left[Q_t^{\max}(s, a) + \sqrt{\frac{2 \log N_t(s)}{N_t(s, a)}} \right]. \quad (32)$$

The tree structure, expansion policy, rollout policy, and exploration term are otherwise identical to Mean-UCT. This condition is a causal diagnostic of the mean-versus-maximum distinction, not a claim that naive maximum backup is universally optimal. A calibrated extreme-bandit allocation rule is a natural subsequent comparison [6].

3.0.2 Evaluation protocol

Each search run uses one persistent tree rooted at s_0 . The algorithm does not commit to a grammar action before the budget is exhausted. No network, learned prior, Dirichlet noise, reward clipping, complexity penalty, or nonlinear optimizer is used. One simulation contains exactly one terminal reward evaluation.

Use the terminal-evaluation budgets

$$\mathcal{B} = \{16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192\}. \quad (33)$$

For each of the four benchmark cells and each search condition, use 100 independent seeds. Run each seed once to $B_{\max} = 8192$ and record all intermediate checkpoints. The same base-seed list is used across algorithms.

3.0.3 Outcomes

Let $\pi_1, \dots, \pi_B$ be the terminal expressions observed by budget B . Define the best-observed reward

$$R_B^{\max} = \max_{1 \leq b \leq B} R(\pi_b), \quad (34)$$

the simple regret

$$r_B = 1 - R_B^{\max}, \quad (35)$$

and the exact-recovery indicator

$$S_B = \mathbf{1} \{R_B^{\max} \geq 1 - 10^{-12}\}. \quad (36)$$

The time to exact recovery is

$$T_{\text{exact}} = \inf \{B : S_B = 1\}. \quad (37)$$

Exact recovery is the primary outcome. Simple regret is secondary because the incorrect decoy in the strong-decoy cells already has simple regret

$$1 - \lambda_{\text{strong}} = 0.003.$$

Thus predictive reward alone cannot distinguish near-perfect approximation from structural recovery. This distinction is consistent with symbolic regression benchmarks that report both predictive and exact-recovery performance [9].

For mechanism diagnosis, record the continue-visit fraction

$$F_B = \frac{N_B(s_0, C)}{N_B(s_0)}, \quad (38)$$

together with the final root recommendation, number of distinct terminal sequences, number of distinct canonical atom sets, and the best reward found through each root action. The final visit-count recommendation is diagnostic only; the primary search output is the best terminal expression encountered.

3.0.4 Hypotheses

Hypothesis 2a: value-semantic effect. Mean-UCT will allocate less search to the continue branch under $\lambda = 0.997$ than under $\lambda = 0.25$, because its exploitation statistic converges toward $V^q(C)$, which lies below the strong-decoy reward. Max-UCT will be less sensitive to this inversion after observing high-reward continuations.

Hypothesis 2b: rarity effect. For every algorithm and fixed λ , exact recovery will be slower for $M = 12$ than for $M = 4$. At moderate budgets, recovery will be organized approximately by the effective rare-event budget

$$B\rho_0. \quad (39)$$

Hypothesis 2c: interaction. The largest recovery difference between Max-UCT and Mean-UCT will occur in the strong-decoy, high-distraction cell

$$(\lambda, M) = (0.997, 12),$$

where the continue branch has both a lower rollout mean than the decoy and a small exact-completion mass.

One pre-specified interaction contrast is

$$\Delta_{\text{semantic}}(B, M) = [P_B^{\text{Max}} - P_B^{\text{Mean}}]_{\lambda=0.997} - [P_B^{\text{Max}} - P_B^{\text{Mean}}]_{\lambda=0.25}, \quad (40)$$

where

$$P_B^A = \Pr_A(S_B = 1)$$

is the exact-recovery probability for algorithm A . The prediction is

$$\Delta_{\text{semantic}}(B, M) > 0,$$

with a larger effect expected at $M = 12$.

3.0.5 Statistical reporting and interpretation

For each budget, cell, and algorithm, report the exact-recovery proportion with a 95% Wilson interval, median simple regret with a seed-bootstrap interval, the empirical survival curve

$$\Pr(T_{\text{exact}} > B),$$

and the continue-visit fraction F_B .

The interpretation is mechanism-based:

- If Mean-UCT fails while Max-UCT succeeds, the mean-return statistic is causally implicated.
- If both methods deteriorate primarily when $M = 12$, completion rarity is the dominant limitation.
- If random search outperforms Mean-UCT in a strong-decoy cell, the tree allocation rule actively suppresses the optimal branch.
- If Mean-UCT succeeds despite the exact value-order inversion, UCT exploration is sufficient at the tested budgets and the mismatch alone is not a complete explanation.
- If Max-UCT performs worse in weak-decoy cells, maximum backup is overoptimistic and should not be treated as a universal replacement for mean backup.
- If simple regret is small while exact recovery remains poor, the experiment demonstrates a separation between predictive approximation and symbolic identification.

3.1 Secondary validation and deferred questions

After the two primary experiments, a nonconfirmatory validation may repeat the key comparison on centered and variance-normalized versions of the motivating quadratic targets, using an additive grammar and direct linear least squares. The original full grammar, nonlinear constant fitting, uniform-prior PUCT, learned policy and value heads, and SRBench evaluation should be introduced only after the controlled mechanisms are understood.

In particular, a subsequent neural experiment should ask whether a learned policy or value function corrects the pure-search failures identified here, or instead learns and amplifies the biased targets produced by mean-return search. Separating that learning question from the present pure-search study is necessary for a causal interpretation.

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